

The Activation Function and the Update Rule

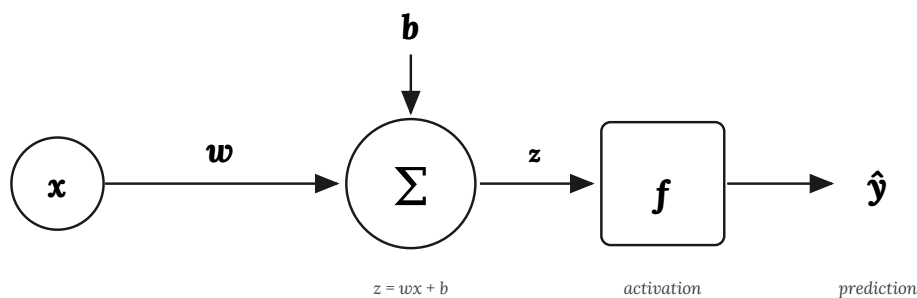
A single perceptron. All work by hand.

SOLUTIONS Every blank from the question paper is filled in. Answers appear in blue. All numerical values are exact to six decimal places.

PART 1 WHAT AN ACTIVATION FUNCTION DOES

1.1 The perceptron

FIGURE 1 – A SINGLE PERCEPTRON



The perceptron does two things in order. First arithmetic, producing z . Then the activation function f , producing the prediction.

$$z = wx + b \quad \hat{y} = f(z)$$

The first part is **arithmetic**. The second part, f , is the **activation function**.

1.2 Why f is needed

Example. A perceptron decides whether a student passes. The input x is the marks obtained. Take $w = 1$ and $b = -40$.

Student	x	$z = wx + b$
A	65	$z = (1)(65) + (-40) = 25$
B	32	$z = (1)(32) + (-40) = -8$

The question asked was *pass or fail*. The required answer is 1 or 0. The arithmetic produced 25 and -8.

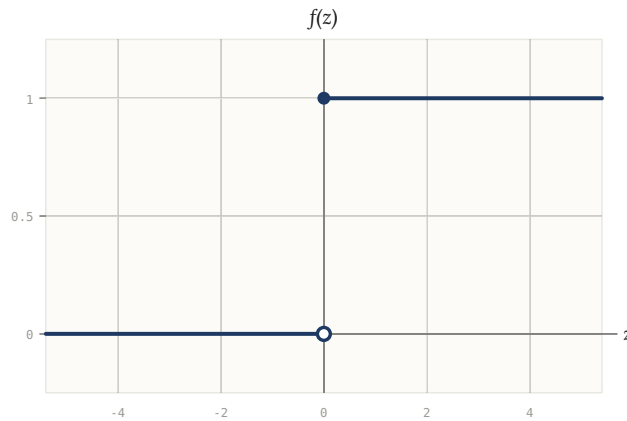
25 is not an answer to the question.

Something must convert z into the answer. That is f .

1.3 The threshold (step) function

$$f(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$$

FIGURE 2 – THE THRESHOLD FUNCTION



The open circle at $(0, 0)$ is excluded; the filled dot at $(0, 1)$ is included. The function jumps at $z = 0$.

Apply it to the two students:

Student	z	$\hat{y} = f(z)$	Decision
A	25	$f(25) = 1$	pass
B	-8	$f(-8) = 0$	fail

1.4 The role of f

For a single perceptron, the activation function f : (i) takes z , which may be any real number, (ii) returns an output in a fixed range, and (iii) so lets the perceptron give an **answer** rather than a measurement.

PART 2 THE UPDATE RULE

2.1 Definitions

$$z = wx + b \quad \hat{y} = f(z) \quad L = \frac{1}{2} (\hat{y} - y)^2$$

y is the target, L is the loss, and α is the learning rate.

2.2 The rule

A large L means a bad prediction. We want L to be **smaller**, so we must change w in whichever direction reduces L . The gradient $\partial L / \partial w$ points in the direction in which L **increases**, so we move against it. This gives the **gradient descent update rule**:

$$w_{\text{new}} = w_{\text{old}} - \alpha \frac{\partial L}{\partial w}$$

2.3 Find $\partial L / \partial w$

w affects z ; z affects $\hat{y}$; $\hat{y}$ affects L . Apply the chain rule.

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial w}$$

Complete each factor.

Since $L = \frac{1}{2}(\hat{y} - y)^2$ and y is a constant

$$\frac{\partial L}{\partial \hat{y}} = \hat{y} - y$$

Since $\hat{y} = f(z)$

$$\frac{\partial \hat{y}}{\partial z} = f'(z)$$

Since $z = wx + b$, and x and b are constants

$$\frac{\partial z}{\partial w} = x$$

Multiply the three factors

$$\frac{\partial L}{\partial w} = (\hat{y} - y) f'(z) x$$

2.4 Substitute into the rule

$$w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f'(z) x$$

Check your answer against the box below. Use this form for every numerical from here on.

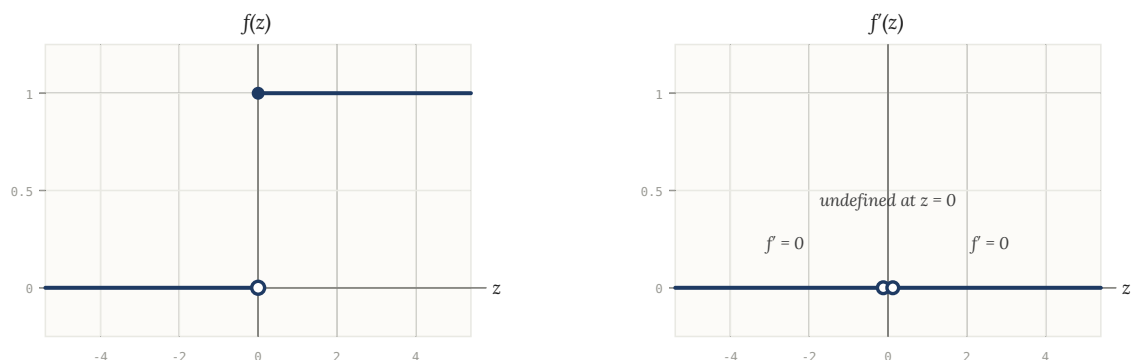
$$w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f'(z) x$$

Three factors control the size of the step: the **error** $(\hat{y} - y)$, the **input** x , and the **derivative of the activation function** $f'(z)$.

PART 3 THE STEP FUNCTION CANNOT LEARN

3.1 Its derivative

FIGURE 3 – THE THRESHOLD FUNCTION AND ITS DERIVATIVE



Both plots use the same vertical scale. The derivative is flat on zero across the whole width. At $z = 0$ the function jumps, so the derivative does not exist there.

For $z > 0$, $f(z) = 1$, a constant

$$f'(z) = 0$$

For $z < 0$, $f(z) = 0$, a constant

$$f'(z) = 0$$

At $z = 0$ the function jumps, so $f'(0)$ is **undefined**. Therefore:

$$\therefore f'(z) = 0 \text{ for all } z \neq 0$$

3.2 Numerical

$$\text{RULE } w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f'(z) x$$

$$\text{GIVEN } x = 2 \quad w = 1 \quad b = 0 \quad y = 0 \quad \alpha = 0.5$$

Weighted sum

$$z = wx + b = (1)(2) + 0 = 2$$

Activation

$$\hat{y} = f(2) = 1 \quad (\text{since } 2 \geq 0)$$

Error

$$\hat{y} - y = 1 - 0 = 1$$

Derivative

$$f'(2) = 0$$

Gradient

$$\frac{\partial L}{\partial w} = (\hat{y} - y) f'(z) x = (1)(0)(2) = 0$$

Update

$$w_{\text{new}} = 1 - (0.5)(0) = 1$$

3.3 Conclusion

The error is 1, the largest it can be. The weight did not change.

The update is zero because

$f'(z) = 0$ at every z where the derivative exists, so the gradient $(\hat{y} - y) f'(z) x$ is $0 \times$ the error, which is 0. The error is measured correctly but it is multiplied away, so it never reaches w .

Increasing α would change nothing, because

α multiplies a gradient that is already zero, and $\alpha \times 0 = 0$ for every value of α . A learning rate can rescale a step, not create one.

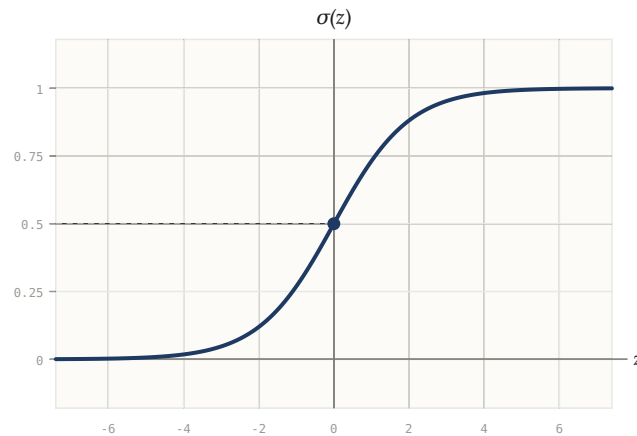
A usable activation function must have a non-zero derivative.

PART 4 THE SIGMOID FUNCTION

4.1 Definition

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

FIGURE 4 – THE SIGMOID FUNCTION



Compare with Figure 2. The sigmoid is **smooth**, so it has a derivative everywhere. It passes through $(0, 0.5)$, and $0 < \sigma(z) < 1$ for every z .

4.2 Derive $\sigma'(z)$

Write $\sigma(z) = (1 + e^{-z})^{-1}$.

Step 1 (given). By the chain rule

$$\sigma'(z) = -(1 + e^{-z})^{-2} \cdot \frac{d}{dz} (1 + e^{-z})$$

Step 2. Evaluate the inner derivative

$$\frac{d}{dz} (1 + e^{-z}) = \underline{-e^{-z}}$$

Step 3. Substitute into Step 1 and simplify

$$\sigma'(z) = \frac{e^{-z}}{(1 + e^{-z})^2}$$

Step 4 (given). Rewrite the numerator using $e^{-z} = (1 + e^{-z}) - 1$

$$\sigma'(z) = \frac{(1 + e^{-z}) - 1}{(1 + e^{-z})^2}$$

Step 5. Split into two fractions

$$\sigma'(z) = \frac{1}{1 + e^{-z}} - \frac{1}{(1 + e^{-z})^2}$$

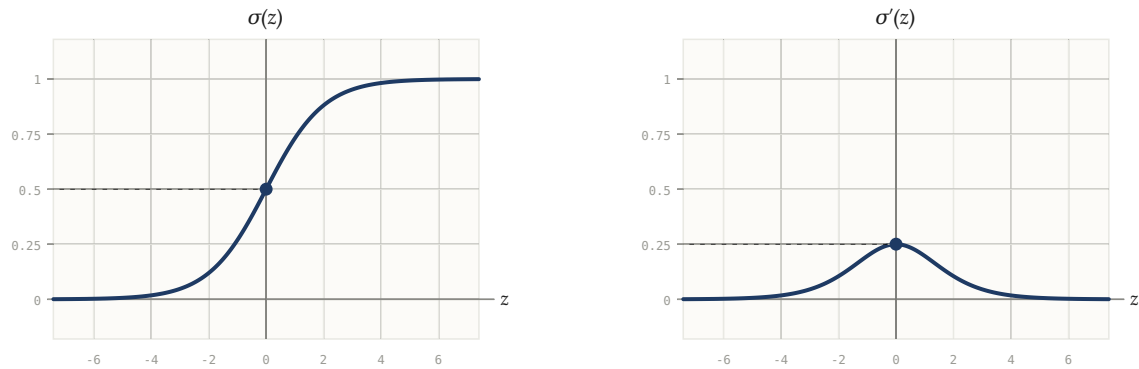
Step 6. Each term is a power of $\sigma(z)$

$$\sigma'(z) = \underline{\sigma(z)} - \underline{\sigma^2(z)} = \underline{\sigma(z) (1 - \sigma(z))}$$

$$\sigma'(z) = \sigma(z) (1 - \sigma(z))$$

4.3 The two graphs

FIGURE 5 – THE SIGMOID AND ITS DERIVATIVE



Both plots use the same vertical scale, 0 to 1. Note where σ' is large and where it is small. Unlike the threshold function, $\sigma'(z)$ is never zero.

PART 5 WORKED EXAMPLE: THE WEIGHT MOVES THE CORRECT WAY

The same numbers as Part 3. **Only f has changed.**

RULE $w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f'(z) x$

GIVEN $x = 2$ $w = 1$ $b = 0$ $y = 0$ $\alpha = 0.5$

Step 1. Weighted sum

$$z = wx + b = (1)(2) + 0 = 2$$

Step 2. Activation

$$\hat{y} = \sigma(2) = \frac{1}{1 + e^{-2}} = \frac{1}{1 + 0.135335} = \frac{1}{1.135335} = 0.880797$$

Step 3. Error

$$\hat{y} - y = 0.880797 - 0 = 0.880797$$

Step 4. Derivative

$$\sigma'(2) = \sigma(2)(1 - \sigma(2)) = (0.880797)(0.119203) = 0.104994$$

Step 5. Gradient

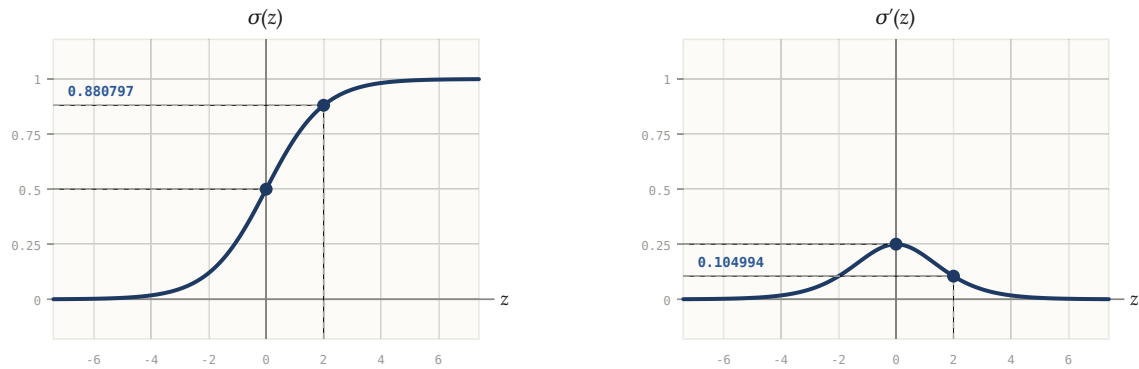
$$\frac{\partial L}{\partial w} = (0.880797)(0.104994)(2) = 0.184956$$

Step 6. Update

$$w_{\text{new}} = 1 - (0.5)(0.184956) = 1 - 0.092478 = 0.907522$$

Rounding: the exact gradient is 0.1849561. Multiplying the displayed six-figure values, $(0.880797)(0.104994)(2)$, returns 0.184957, which differs in the last place only. $w_{\text{new}} = 0.907522$ either way.

FIGURE 6 – THE SAME TWO GRAPHS, WITH $z = 2$ MARKED



At $z = 2$ the sigmoid is already flattening, so $\sigma'(2) = 0.104994$ has fallen well below its peak. Read both values off the dashed guides.

Direction check

Requirement	Consequence
Target is $y = 0$	$\hat{y}$ must decrease
$\hat{y} = \sigma(z)$ increases with z	z must decrease
$z = wx + b$ with $x = 2 > 0$	w must decrease
Result	$w : 1 \rightarrow 0.907522 \quad \checkmark$

Comparison with Part 3

IDENTICAL x , w , b , y AND A . ONLY $f'(z)$ DIFFERS.

Activation	$f'(2)$	$\partial L / \partial w$	w_{new}
Threshold	0	0	1 (no change)
Sigmoid	0.104994	0.184956	0.907522

PART 6 PRACTICE

Show every step. Give all values to six decimal places.

RULE $w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f'(z) x$

GIVEN $x = 3 \quad w = 0.5 \quad b = -1 \quad y = 1 \quad \alpha = 2$

USEFUL $e^{-0.5} = 0.606531$

Step 1. Weighted sum

$$z = wx + b = \underline{(0.5)(3) + (-1) = 1.5 - 1 = 0.5}$$

Step 2. Activation

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + 0.606531} = \frac{1}{1.606531} = 0.622459$$

Step 3. Error

$$\hat{y} - y = \underline{0.622459 - 1 = -0.377541}$$

Step 4. Derivative

$$\sigma'(z) = \sigma(z)(1 - \sigma(z)) = \underline{(0.622459)(0.377541) = 0.235004}$$

Step 5. Gradient

$$\frac{\partial L}{\partial w} = (\hat{y} - y) \sigma'(z) x = \underline{(-0.377541)(0.235004)(3) = -0.266170}$$

Step 6. Update

$$w_{\text{new}} = \underline{0.5 - (2)(-0.266170) = 0.5 + 0.532341 = 1.032341}$$

Rounding: the exact gradient is -0.2661704 , so $(2)(0.2661704) = 0.5323408$ and $w_{\text{new}} = 1.032341$. Doubling the rounded -0.266170 instead gives 1.032340 , one unit low in the sixth place. Either answer is acceptable; carry the seventh digit if you want the printed value.

Step 7 — state the direction of the change and justify it in one line.

w **increases**, from 0.5 to 1.032341.

The target $y = 1$ lies above the prediction $\hat{y} = 0.622459$, so $\hat{y}$ must rise; σ rises with z and $x = 3$ is positive, so w must rise. Equivalently, the gradient is negative, and subtracting a negative quantity adds to w .

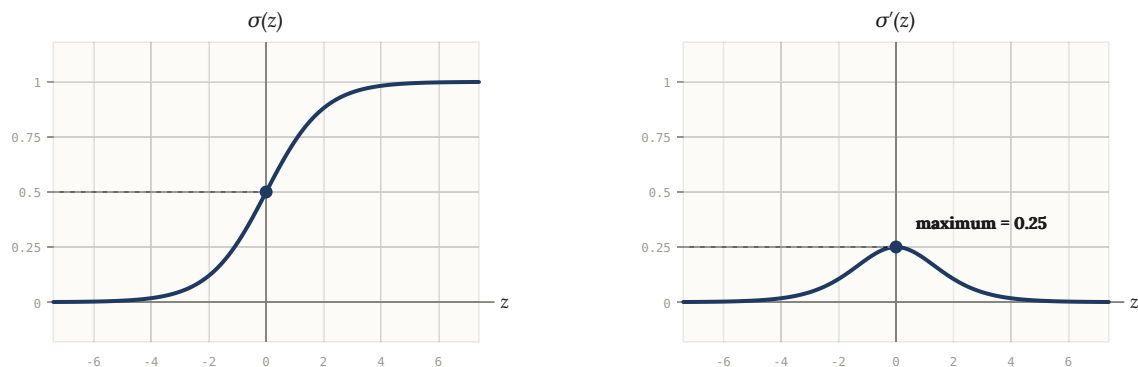
ANSWERS, FOR CHECKING

$z = 0.5$ | $\hat{y} = 0.622459$ | $\hat{y} - y = -0.377541$ | $\sigma'(z) = 0.235004$ | $\partial L / \partial w = -0.266170$ | $w_{\text{new}} = 1.032341$

PART 7 THE LARGEST SIGMOID GRADIENT, READ FROM THE GRAPH

7.1 Read the maximum off the curve

FIGURE 7 — KEY FIGURE. READ THE PEAK HEIGHT FROM THE RIGHT-HAND PLOT



Both vertical axes run from 0 to 1 with identical marks, so the two curves can be compared directly. The dashed guides on the right-hand plot locate the peak.

Fill in by reading Figure 7.

The peak of $\sigma'(z)$ occurs at

$$z = \underline{0}$$

The height of that peak is

$$\sigma'_{\text{max}} = \underline{0.25}$$

On the left-hand plot, the value of $\sigma(z)$ at that same z is

$$\sigma(z) = \underline{0.5}$$

$$\sigma'(z) \leq 0.25 \text{ for every } z$$

Confirm with the formula: at $z = 0$ we have $\sigma(0) = 0.5$, so $\sigma'(0) = (0.5)(1 - 0.5) = 0.25$. ✓

7.2 The problem

At **every** value of z , the sigmoid multiplies the gradient by at most 0.25. The numerical below shows what that costs even in the best possible case, where $z = 0$ and the derivative takes its largest value.

$$\text{RULE } w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f'(z) x$$

$$\text{GIVEN } x = 1 \quad w = 0 \quad b = 0 \quad y = 1 \quad \alpha = 1$$

Weighted sum

$$z = wx + b = (0)(1) + 0 = 0$$

Activation

$$\hat{y} = \sigma(0) = \frac{1}{1 + e^0} = \frac{1}{2} = 0.5$$

Error

$$\hat{y} - y = 0.5 - 1 = -0.5$$

Derivative

$$\sigma'(0) = (0.5)(1 - 0.5) = 0.25 \quad \leftarrow \text{the largest value it can ever take}$$

Gradient

$$\frac{\partial L}{\partial w} = (-0.5)(0.25)(1) = -0.125$$

Update

$$w_{\text{new}} = 0 - (1)(-0.125) = 0.125$$

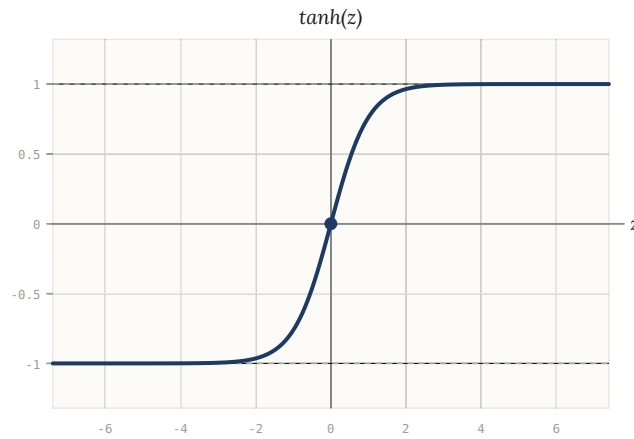
This is the sigmoid at its **best possible** point, and it still shrinks the gradient to one quarter. The step taken is only 0.125.

PART 8 THE TANH FUNCTION

8.1 Definition

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}} \quad -1 < \tanh(z) < 1$$

FIGURE 8 – THE TANH FUNCTION



Compare with Figure 4. The same S-shape, but the output range is -1 to 1 , and the curve passes through the origin.

8.2 Its derivative

Let $u = e^z - e^{-z}$ and $v = e^z + e^{-z}$, so $\tanh = u / v$.

Step 1 (given). Differentiate each part

$$u' = e^z + e^{-z} = v \quad v' = e^z - e^{-z} = u$$

Step 2. Quotient rule, then substitute $u' = v$ and $v' = u$

$$\tanh'(z) = \frac{u'v - uv'}{v^2} = \frac{v^2 - u^2}{v^2}$$

Step 3. Split the fraction and simplify

$$\tanh'(z) = 1 - \frac{u^2}{v^2} = 1 - \tanh^2(z)$$

The two parts exchange with each other. That is what makes this derivation short.

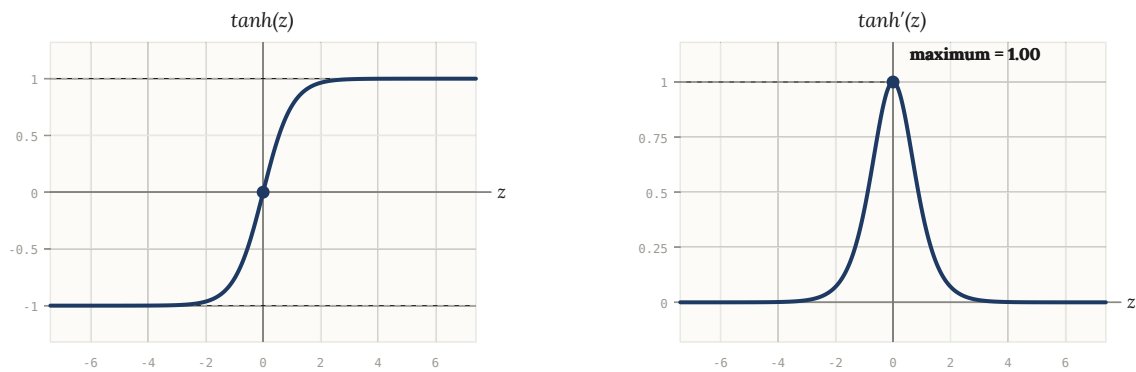
$$\tanh'(z) = 1 - \tanh^2(z)$$

At $z = 0$, since $\tanh(0) = 0$

$$\tanh'(0) = 1 - (0)^2 = \underline{1}$$

8.3 The two graphs

FIGURE 9 – TANH AND ITS DERIVATIVE



The derivative reaches a peak of 1.00 at $z = 0$, compared with 0.25 for the sigmoid in Figure 7. Its vertical axis runs 0 to 1, the same as Figure 7, so the two peaks are directly comparable.

8.4 The same numerical, with tanh

RULE $w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f'(z) x$

GIVEN, UNCHANGED FROM 7.2 $x = 1$ $w = 0$ $b = 0$ $y = 1$ $\alpha = 1$

Weighted sum

$$z = wx + b = (0)(1) + 0 = 0$$

Activation

$$\hat{y} = \tanh(0) = 0$$

Error

$$\hat{y} - y = 0 - 1 = -1$$

Derivative

$$\tanh'(0) = 1 - (0)^2 = 1$$

Gradient

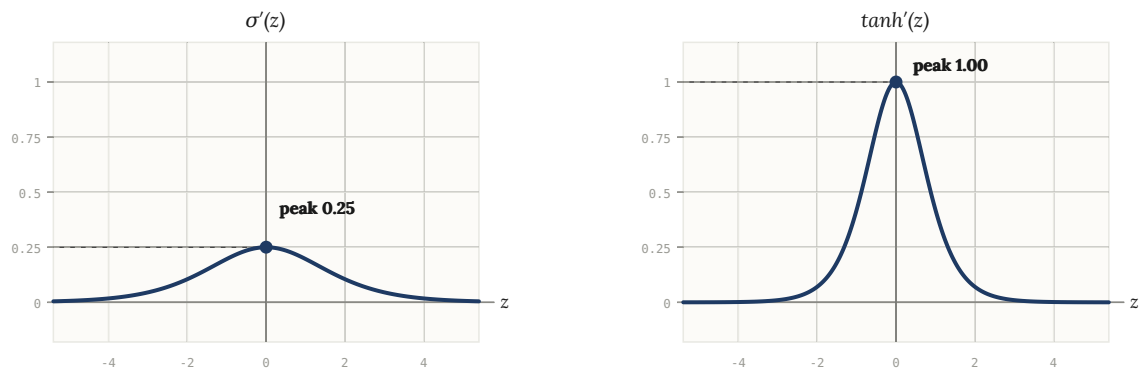
$$\frac{\partial L}{\partial w} = (-1)(1)(1) = -1$$

Update

$$w_{\text{new}} = 0 - (1)(-1) = 1$$

8.5 Comparison

FIGURE 10 – THE TWO DERIVATIVES ON IDENTICAL AXES



Identical horizontal and vertical scales on both plots. The difference in peak height is the whole point: 0.25 against 1.00.

	$\hat{y}$	$\hat{y} - y$	$f'(0)$	$\partial L / \partial w$	$w : 0 \rightarrow$	Step size
Sigmoid	0.5	-0.5	0.25	-0.125	0.125	0.125
Tanh	0	-1	1	-1	1	1

$$\frac{\text{tanh step}}{\text{sigmoid step}} = \frac{1}{0.125} = 8$$

Account for the factor of 8. Two quantities changed.

$$\text{error ratio} = \frac{1}{0.5} = \underline{2} \quad \text{derivative ratio} = \frac{1}{0.25} = \underline{4}$$

$$\underline{2} \times \underline{4} = 8$$

SUMMARY THE THREE ACTIVATION FUNCTIONS

Activation	$f(z)$	$f'(z)$	max f'	Learns?
Threshold	1 if $z \geq 0$, else 0	0	0	No
Sigmoid	$1 / (1 + e^{-z})$	$\sigma(z)(1 - \sigma(z))$	0.25	Yes, slowly
Tanh	$(e^z - e^{-z}) / (e^z + e^{-z})$	$1 - \tanh^2(z)$	1	Yes, faster

In the update rule

$$w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f'(z) x$$

the factor $f'(z)$ decides **whether** learning happens at all, and **how fast**.

In one line: why can a perceptron using the threshold function never be trained by this rule?

Because $f'(z) = 0$ at every z where it exists, the whole update term $\alpha(\hat{y} - y) f'(z) x$ is zero, so w never moves however wrong the prediction is, and no choice of α can repair it.

All numerical values in this worksheet are exact to six decimal places. Every plot is drawn to scale, and all derivative plots share a vertical axis running from 0 to 1 so that peak heights may be compared directly.